

Troubleshooting & Monitoring

Lab Activity: Diagnose a Mystery Host

Overview

This hands-on lab gives you the opportunity to use the core diagnostic tools covered in Unit 7 in a realistic troubleshooting scenario. You will run the commands on a terminal while observing, and trying to understanding the output.

Duration	20 to 30 minutes
Format	Instructor-led live demo with class discussion
Tools Needed	Linux or macOS terminal (or Windows with WSL installed)
Prerequisites	Students should have completed the Unit 7 lecture slides

Scenario

Tell students: *"You are a network administrator. A user has reported that they cannot reach a remote server. Your job is to investigate the problem systematically, starting from the most basic check and working up through the network layers."*

Step-by-Step activity

1**Test Basic Reachability (ping)****Command:** `ping google.com`

Run ping for a few seconds, then stop it. Ask students: Is the host alive? What is the average RTT? Is there any packet loss?

2**Trace the Path (tracert)****Command:** `tracert google.com`

Walk through each hop with students. Point out any hop showing * * * and ask: What could cause no response at this hop?

3**Check Active Connections (netstat)****Command:** `netstat -tunapn | grep LISTEN`

Ask students to identify which services are currently listening and on which ports.

4

Inspect the Network Interface (ifconfig)

Command: `ifconfig` (or: `ip addr show`)

Ask students to locate the machine IP address, the subnet mask, and the MAC address.

5

Read a System Log

Command: `tail -20 /var/log/syslog` (or: `journalctl -n 20`)

Point out the timestamp format, severity levels (ERROR, WARNING, INFO), and any notable entries.

Discussion Questions

After completing the activity, use these questions to check your understanding:

- At which step would you first suspect a routing problem?
- If ping succeeds but the application still fails, which layer do you investigate next?
- What log file would you check after a failed SSH login attempt?
- What does a * * * output in traceroute tell you about that hop?
- How can you use netstat to confirm whether a web server is running on port 80?

Optional Challenge

Upon completion of this activity, run ping against a non-existent IP address and compare the output to a successful ping:

```
ping 192.168.99.99
```

You should be able to explain:

- Why there is no reply from the target
- What 100% packet loss indicates about reachability
- How this output differs from a host that is online but blocking ICMP

Expected Learning Outcomes

By the end of this lab, you should be able to:

1. Use ping to test host reachability and measure round-trip latency
2. Use traceroute to identify where along a path a failure or delay occurs
3. Use netstat to list active connections and listening ports
4. Use ifconfig to read network interface configuration details
5. Read and interpret basic system log entries for troubleshooting purposes

6. Apply a layer-by-layer approach when diagnosing connectivity issues

Instructor Notes

- No special lab environment is required. Any machine with a terminal will work.
- If network access is restricted, substitute a known internal IP for google.com in ping and traceroute.
- Predict the output before each command is run.
- The optional challenge works well as a pair activity if time allows.
- You can run all five commands on your own machines and interpret the output.